

CHAPTER 4: CHEMICAL BONDING & MOLECULAR STRUCTURE (JEE FOCUS)

Topics Covered:

- Ionic bond
- Covalent bond
- Lewis structures
- VSEPR theory
- Hybridization
- Molecular Orbital Theory
- Bond parameters

Core Concepts:

- Ionic bond → electron transfer
- Covalent bond → electron sharing
- Lattice energy → stability of ionic compounds
- Lewis structure → octet rule
- VSEPR → minimize repulsion
- Shape depends on lone pairs + bond pairs
- Hybridization → mixing of orbitals
- sp , sp^2 , sp^3 → geometry fixed
- Bond order → stability measure
- MOT → bonding & antibonding orbitals
- Magnetic nature → unpaired electrons

Key Formulas:

- Bond order = $(N_b - N_a)/2$
- Dipole moment = $q \times r$
- Formal charge = valence – (lone + $1/2$ bonding)

Important Patterns:

- More bond order → more stability
- More lone pairs → bond angle decreases
- sp → linear, sp^2 → trigonal, sp^3 → tetrahedral
- Bent shapes due to lone pairs
- MOT → O_2 paramagnetic (important)
- Resonance → lowers energy
- Polar vs non-polar → symmetry check

Common Mistakes:

- Ignoring lone pairs in shape
- Wrong hybridization
- Forgetting resonance
- Wrong bond order calculation
- Confusing geometry vs shape
- Ignoring symmetry in polarity

Examples (Important Cases):

- Shape of NH_3 , H_2O
- Hybridization of CH_4 , C_2H_4 , C_2H_2
- MOT of O_2 , N_2
- Bond order comparison
- Polarity of CO_2 , H_2O

Exam Tips:

- Draw structure before answering
- Count electron pairs carefully
- Memorize hybridization table
- Use symmetry for polarity
- Practice MOT diagrams
- Focus on shape-based questions
- High weight chapter → revise multiple times